Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A point on the ellipse $4x^2 + 9y^2 = 36$. Where the normal is parallel to the line $4x - 2y - 5 = 0$ is:

(A) $\left(\frac{9}{5}, \frac{8}{5}\right)$

(B) $\left(\frac{8}{5}, -\frac{9}{5}\right)$

(C) $\left(-\frac{9}{5}, \frac{8}{5}\right)$

(D) $\left(\frac{8}{5}, \frac{9}{5}\right)$

Q.2

If an ellipse slides between two perpendicular straight lines, then the locus of its centre is

(A) a parabola

(B) an ellipse

(C) a hyperbola

(D) a circle

Q.3

If p is the length of the perpendicular from the focus S of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ to a tangent at a point P on the ellipse, then $\frac{2\mathrm{a}}{\mathrm{SP}} - 1 =$

(A) $\frac{\mathrm{a}^2}{\mathrm{p}^2}$

(B) $\frac{b^2}{p^2}$

(C) p^2

(D) $\frac{\mathrm{a}^2+\mathrm{b}^2}{\mathrm{p}^2}$

Q.4

The sum of the squares of the perpendiculars on any tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ from two points on the minor axis each at a distance $\sqrt{a^2 - b^2}$ from the centre is

(A) $2\mathrm{a}^2$

(B) $2\,\mathrm{b}^2$

(C) $a^2 + b^2$

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(D) $a^2 - b^2$

Q.5

The equations of the tangents of the ellipse $\frac{x^2}{16} + \frac{y^2}{45} = 1$ which passes through the point $(6,$ 10 is

(A) $11x - 2y + 46 = 0, x - 2y + 14 = 0$

(B) $11x - 2y - 46 = 0, x - 2y - 14 = 0$

(C) $11x - 2y + 46 = 0, x - 2y - 14 = 0$

(D) $11x - 2y - 46 = 0, x - 2y + 14 = 0$

Q.6

$\frac{x}{a} + \frac{y}{b} = \sqrt{2}$ touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then its eccentric angle θ is equal to

(A) 0

(B) 90°

(C) 45°

(D) 60°

Q.7

The distance of the center of the ellipse $x^2 + 2y^2 - 2 = 0$ to those tangents of the ellipse which are equally inclined to both the axes is

(A) $\frac{3}{\sqrt{2}}$

(B) $\frac{\sqrt{3}}{2}$

(C) $\frac{\sqrt{2}}{3}$

(D) $\sqrt{\frac{3}{2}}$

Q.8

If $(\sqrt{3})bx + ay = 2ab$ touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then eccentric angle of the poin contact is

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{2}$

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Q.9

The line $3x + 5y = k$ touches the ellipse $16x^2 + 25y^2 = 400$ if $\mathbf{k}$ is

(A) $\pm\sqrt{5}$

(B) $\pm\sqrt{15}$

(C) ± 25

(D) None of these

Q.10

If the ellipse $9\mathrm{x}^2 + 16\mathrm{y}^2 = 144$ intercepts the line $3x + 4y = 12$, then what is the length of the chord so formed?

(A) 5 units

(B) 6 units

(C) 8 units

(D) 10 units

Q.11

Consider any point P on the ellipse $\frac{\mathrm{x}^2}{25} + \frac{\mathrm{y}^2}{9} = 1$ in the first quadrant. Let r and s represent its distances from $(4, 0)$ and $(-4, 0)$ respectively, then $(\mathrm{r} + \mathrm{s})$ is equal to

(A) 10 unit

(B) 9 unit

(C) 8 unit

(D) 6 unit

Q.12

Let $\mathrm{S}_1,\ \mathrm{S}_2$ be the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{8} = 1$. If $A(x + y)$ is any point on the ellipse, then the maximum area of the triangle $\mathrm{AS}_1\ \mathrm{S}_2(\mathrm{in}$ square units) is

(A) $2\sqrt{2}$

(B) $2\sqrt{3}$

(C) 8

(D) 4

Q.13

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Let E be the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ and C be the circle $x^2+y^2=9$. Let $P=(1,2)$ and $Q=(2,1)$. Which one of the following is correct?

(A) Q lies inside C but outside E

(B) Q lies outside both C and E

(C) P lies inside both C and E

(D) P lies inside C but outside E.

Q.14

The line passing through the extremity A of the major axis and the extremity B of the minor axis of the ellipse $x^2+9y^2=9$ meets its auxiliary circle at the point M. Then the area of the triangle with vertices A, M and the origin O is

(A) $\frac{31}{10}$

(B) $\frac{29}{10}$

(C) $\frac{21}{10}$

(D) $\frac{27}{10}$

Q.15

The locus of the point of intersection of two tangents of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ which are inclined at angles θ_1, and θ_2 with the major axis such that $\tan^2\theta_1+\tan^2\theta_2$ is constant, is

(A) $4x^2y^2+2\left(x^2-a^2\right)\left(y^2-b^2\right)=k\left(x^2-a^2\right)^2$

(B) $4x^2y^2-2\left(x^2-a^2\right)\left(y^2-b^2\right)=k\left(x^2-a^2\right)^2$

(C) $4x^2y^2-2\left(x^2-a^2\right)\left(y^2-b^2\right)=k\left(x^2+a^2\right)^2$

(D) None of these

Answers & Solutions

Q.1 Answer:

$\left(\frac{9}{5}, \frac{8}{5}\right)$

Solution:

Slope of the normal is 2, so slope of the tangent is $-\frac{1}{2}$.

Equation of the tangent is $y=-\frac{1}{2}x \pm \sqrt{9\times\frac{1}{4}+4}$

$\Rightarrow \quad x+2y=\pm 5$ **...(1)**

If it touches at (x_1, y_1), then equation of the tangent at (x_1, y_1) to the ellipse is

$\frac{xx_1}{9}+\frac{yy_1}{4}=1$ **...(2)**

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Comparing (1) and (2) we get

$$\frac{x_1}{9} = \frac{y_1}{8} = \pm\frac{1}{5}$$

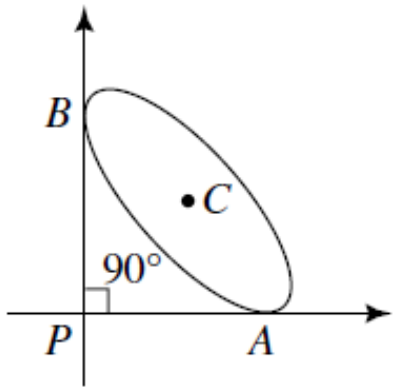
$$\Rightarrow \quad (x_1, y_1) = \left(\frac{9}{5}, \frac{8}{5}\right)$$

Q.2 Answer:

a circle

Solution:

Let $2a, 2b$ be the length of the major and minor axes respectively of the ellipse. If the ellipse slides between two perpendicular lines, the point of intersection P of these lines being the point of intersection of perpendicular tangents lies on the Director circle of the ellipse. This means that the centre C of the ellipse is always at a constant distance $\sqrt{a^2+b^2}$ from P. Hence the locus of C is a circle.



Q.3 Answer:

$\frac{b^2}{p^2}$

Solution:

Let the point P **be** $(\mathrm{a}\cos\theta, \mathrm{b}\sin\theta)$

The tangent at P **is** $\frac{\mathrm{x}}{\mathrm{a}}\cos\theta + \frac{\mathrm{y}}{\mathrm{b}}\sin\theta = 1$ **....(i)**

The perpendicular distance p **of** $\mathrm{S}(\mathrm{ae}, 0)$ **form (i) is given by** $\mathrm{p}^2 = \frac{(\mathrm{e}\cos\theta - 1)^2}{\frac{\cos^2\theta}{\mathrm{a}^2} + \frac{\sin^2\theta}{\mathrm{b}^2}}$

$$\Rightarrow \frac{1}{p^2} = \frac{\frac{\cos^2\theta}{a^2} + \frac{\sin^2\theta}{b^2}}{(e\cos\theta - 1)^2}$$

$$\Rightarrow \frac{b^2}{p^2} = \frac{\frac{b^2}{a^2}\cos^2\theta + 1 - \cos^2\theta}{(e\cos\theta - 1)^2}$$

$$= \frac{\left(\frac{b^2}{a^2} - 1\right)\cos^2\theta + 1}{(e\cos\theta - 1)^2}$$

$$= \frac{1 - e^2\cos^2\theta}{(\mathrm{e}\cos\theta - 1)^2} = \frac{1 + \mathrm{e}\cos\theta}{1 - \mathrm{e}\cos\theta}$$

Now $\mathrm{SP} = \mathrm{a}(1 - \mathrm{e}\cos\theta)$

$$\therefore \frac{2\mathrm{a}}{\mathrm{SP}} - 1 = \frac{2\mathrm{a}}{\mathrm{a}(1 - \mathrm{e}\cos\theta)} - 1 = \frac{1 + \mathrm{e}\cos\theta}{1 - \mathrm{e}\cos\theta} = \frac{\mathrm{b}^2}{\mathrm{p}^2}$$

Q.4 Answer:

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$2a^2$

Solution:

$\sqrt{a^2-b^2}=\pm ae(0,ae)$

So, the points are $(ae,0)$ **and** $(0,-ae)$.

Let $\frac{x}{a}\cos\theta+\frac{y}{b}\sin\theta=1$ **be a tangent then sum of squares of perpendicular from these points is**

$$=\frac{\left(1-\frac{ae}{b}\sin\theta\right)^2+\left(1+\frac{ae}{b}\sin\theta\right)^2}{\frac{\cos^2\theta}{a^2}+\frac{\sin^2\theta}{b^2}}$$

$$=\frac{2\left(1+\frac{a^2e^2}{b^2}\sin^2\theta\right)}{\frac{\cos^2\theta}{a^2}+\frac{\sin^2\theta}{b^2}}$$

$$=2a^2\left(\frac{b^2+\left(a^2-b^2\right)\sin^2\theta}{b^2(1-\sin\theta)+a^2\sin^2\theta}\right)=2a^2$$

Q.5 Answer:

$11x-2y-46=0, x-2y+14=0$

Solution:

$y=mx\pm\sqrt{16m^2+45}$

$\Rightarrow(10-6m)=\pm\sqrt{16m^2+45}\Rightarrow m=\frac{11}{2},\frac{1}{2}$

Equation of tangents are $11x-2y-46=0$ **and** $x-2y+14=0$

Q.6 Answer:

45°

Solution:

Tangent at $(a\cos\theta,b\sin\theta)$ **to the ellipse is** $\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$

Comparing with

$\frac{x}{a\sqrt{2}}+\frac{y}{b\sqrt{2}}=1\Rightarrow\cos\theta=\frac{1}{\sqrt{2}},\sin\theta=\frac{1}{\sqrt{2}}$

$\Rightarrow\theta=45^\circ$

Q.7 Answer:

$\sqrt{\frac{3}{2}}$

Solution:

Equation of ellipse is $\frac{x^2}{2}+\frac{y^2}{1}=1$. **General tangent to the ellipse of slope** m **is**

$y=mx\pm\sqrt{2m^2+1}$

Since this is equally inclined to axes, so $m=\pm1$. **Thus tangents are** $y=\pm x\pm\sqrt{2+1}$

$=\pm x\pm\sqrt{3}$

Distance of any of these tangent these from origin is equal to $\sqrt{\frac{3}{2}}$

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Q.8 Answer:

$\frac{\pi}{6}$

Solution:

Equation of tangent is $\frac{x\sqrt{3}}{a}+\frac{y}{b}\frac{1}{2}=1$ **and equation of tangent at the point** $(a\cos\phi, b\sin\phi)$ **is** $\frac{x}{a}\cos\phi+\frac{y}{b}\sin\phi=1$. **Both are same i.e.,** $\cos\phi=\frac{\sqrt{3}}{2}, \sin\phi=\frac{1}{2}\Rightarrow\phi=\frac{\pi}{6}$

Q.9 Answer:

± 25

Solution:

Putting $5y=\mathbf{k}-3x$ **in the equation of the ellipse** $16x^2+25y^2=400$ **we get** $16x^2+(k-3x)^2=400$

$\Rightarrow 25x^2-6\mathbf{k}x+k^2-400=0$ **Now** $\mathrm{D}=0$

$\Rightarrow 36k^2-100\left(k^2-400\right)=0\Rightarrow k=\pm 25$

Q.10 Answer:

5 units

Solution:

(a) Here,

$9x^2+16y^2=144$ **and** $3x+4y=12$

$\Rightarrow x=\frac{12-4y}{3}$

So, $9\left(\frac{12-4y}{3}\right)^2+16y^2=144$

On solving we get, $y=0,3$ **For** $y=0; x=4$

For $y=3; x=0$

$\Rightarrow$ **Length of chord** $=\sqrt{(0-3)^2+(4-0)^2}=\sqrt{9+16}$

$=\sqrt{25}=5$ **units**

Q.11 Answer:

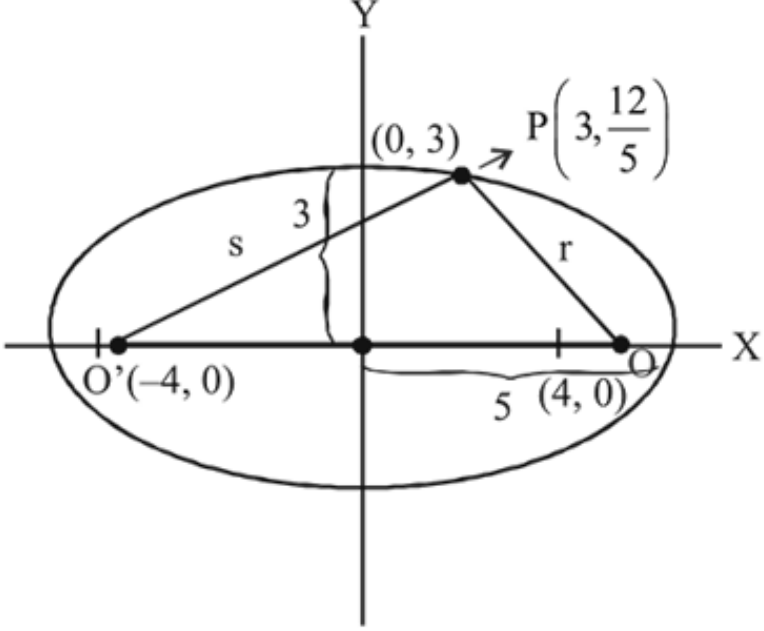
10 unit

Solution:

$\frac{x^2}{25}+\frac{y^2}{9}=1$

Put $x=3$

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$\frac{9}{25} + \frac{y^2}{9} = 1$
$y = \frac{12}{5}$
$P = (3, 12/5)$

$$r = PO = \sqrt{(4-3)^2 + \left(0 - \frac{12}{5}\right)^2}$$
$$= 17/5$$
$$S = PO' = \sqrt{[-4-3] + \left(0 - \frac{12}{5}\right)^2}$$
$$= 33/5$$

$r + s = \frac{17}{5} + \frac{33}{5} = \frac{50}{5} = 10$ unit

Q.12 Answer:

8

Solution:

Equation of ellipse is $\frac{x^2}{16} + \frac{y^2}{8} = 1$
where, $a = 4, b = 2\sqrt{2}$
Eccentricity, $e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{8}{16}} = \frac{1}{\sqrt{2}}$
Area is maximum when vertex is $(0,\ b)$
- Maximum area $= \frac{1}{2} \times 2ae \times b$
$= \frac{1}{2} \times 2 \times 4 \times 2\sqrt{2} \times \frac{1}{\sqrt{2}} = 8$ **sq. units**

Q.13 Answer:

P lies inside C but outside E.

Solution:

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Given equation of ellipse E is $\frac{x^2}{9}+\frac{y^2}{4}=1$

$\Rightarrow \frac{4x^2+9y^2}{36}=1 \Rightarrow 4x^2+9y^2=36$

$\Rightarrow 4x^2+9y^2-36=0$

And C : eqn of circle is $x^2+y^2=9$ Which can be rewritten as

$x^2+y^2-9=0$

For a point $P(1,2)$ we have $4(1)^2+9(2)^2-36=40-36>0[$ from

and $1^2+2^2-9=5-9<0$ [from

Point P lies outside of E and incide of r

Q.14 Answer:

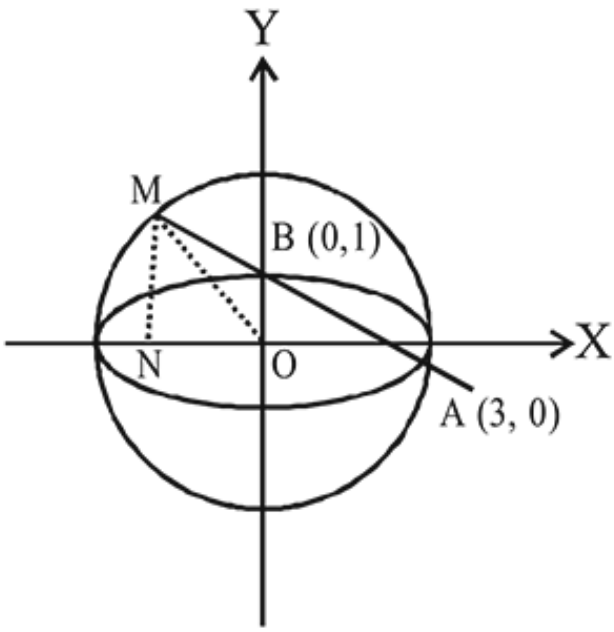
$\frac{27}{10}$

Solution:

Equation of the ellipse is $\frac{x^2}{9}+\frac{y^2}{1}=1$ An end of the major axis A be say $(3,0)$ and an end of the minor axis B be say $(0,1)$. Equation of AB is therefore.

$\frac{x}{3}+\frac{y}{1}=1$

Y



Equation of the auxiliary circle is $x^2+y^2=9 \ldots .$

Solving the equation (1) and (2) we get $x^2+\left(1-\frac{x}{3}\right)^2=9 \Rightarrow x^2+1+\frac{x^2}{9}-\frac{2x}{3}=9$

$\Rightarrow 5x^2-3x-36=0 \Rightarrow (5x+12)(x-3)=0$

$x=-\frac{12}{5} \Rightarrow y=1-\frac{1}{3}\left(-\frac{12}{5}\right)=\frac{9}{5}$

$\therefore$ Coordinates of M are $\left(-\frac{12}{5},\frac{9}{5}\right)$

Area of $\Delta AOM=\frac{1}{2}.OA \cdot MN=\frac{1}{2}\times 3\times\frac{9}{5}=\frac{27}{10}$

Q.15 Answer:

$4x^2y^2-2\left(x^2-a^2\right)\left(y^2-b^2\right)=k\left(x^2-a^2\right)^2$

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Solution:

If the tangent $y = mx \pm \sqrt{a^2m^2 + b^2}$ **passes through** (x_1, y_1) **then** $(y_1 - mx_1)^2 = a^2m^2 + b^2$

or $\left(\mathrm{x}_1^2 - \mathrm{a}^2\right) \mathrm{m}^2 - 2\mathrm{x}_1\mathrm{y}_1\, \mathrm{m} + \mathrm{y}_1^2 - \mathrm{b}^2 = 0$**. If**

the roots be m_1 **and** m_2 **then** $m_1 + m_2 = \frac{2x_1y_1}{x_1^2 - a^2}$ **and** $m_1m_2 = \frac{y_1^2 - b^2}{x_1^2 - a^2}$

Given : $\tan^2\theta_1 + \tan^2\theta_2 =$ **constant** $= \mathrm{k}$ **(say)**

$\Rightarrow \mathrm{m}_1^2 + \mathrm{m}_2^2 = \mathrm{k}$ **or** $\left(\mathrm{m}_1 + \mathrm{m}_2\right)^2 - 2\,\mathrm{m}_1\,\mathrm{m}_2 = \mathrm{k}$

$\Rightarrow \frac{4x^2y^2}{\left(x_1^2 - a^2\right)^2} - \frac{2\left(y_1^2 - b^2\right)}{\left(x_1^2 - a^2\right)} = k$

or

$4x_1^2y_1^2 - 2\left(x_1^2 - a^2\right)\left(y_1^2 - b^2\right) = k\left(x_1^2 - a^2\right)^2$

Hence locus of P **is**

$4x^2y^2 - 2\left(x^2 - a^2\right)\left(y^2 - b^2\right) = k\left(x^2 - a^2\right)^2$

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